General Architecture of the Multi-Agent Instructional Design System

Input

Course goals & objectives
- learning goals
- duration
- learning setting

Learners' profile
- prior knowledge
- level, age
- class size

Learning resources
- labs, LMS, tools
- technology
- language

Small language model
- open-weight model
- self-hosted
- no external API

1. Instructional Design & Planning Layer (multi-agent)

1. **Needs Analysis** — gaps, topics, constraints
2. **Outcome Designer** — objectives, modules
3. **Assessment Designer** — blueprint, items, rubrics
4. **Activity Designer** — learning activities
5. **Content Designer** — media, materials
6. **Implementation Planner** — sessions, guides, manuals

Backward Design order

Developer agents: items, activities, materials, sessions per module

2. Small Language Model Support Layer (orchestration)

- **Alignment Graph** — links objectives, activities, assessments
- **Validation** — checks each submitted artifact
- **Repair Loop** — errors back to the owner agent
- **Schema per Artifact** — constrained decoding (vLLM)
- **Audit & Tracking** — graph and violation ledger

3. Teaching & Learning Process Layer

dashed: not built or measured in this study

- **Web App** — teacher traces and reviews the design
- **Delivery** — teaches with session plans
- **Students** — study, practise, take assessments
- **Interaction** — teacher and students
- **Learning outcomes** — knowledge, skills

Output

Course design
- objectives, modules
- assessment plan
- items, rubrics

Teaching materials
- learning activities
- media selection
- materials per module

Delivery plan
- session plans
- instructor guide
- operator manual

Audit record
- alignment graph
- violation ledger
- traced in web app

Continuous Improvement (Feedback Loop) — future extension

Teacher edits → trace affected artifacts in the graph → owner agent revises → re-check